



Molecular biology lab

Iman Abunar | | Gene family

Jordan University of Science & Technology
Department of Biotechnology & Genetic Engineering
Molecular Biology lab BT 453

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- ***Reference:***

1. lab notes.
2. Slides on E-learning.
3. Lab Manual.
4. Recommended website:
 1. Wikipedia.com
 2. qiagen.com
 3. dev.biologists.org
 4. science.sciencemag.org
 5. worldwide.promega.com



MOLECULAR LAB

You will learn principles and practice of basic bacterial culture techniques, Agarose gel electrophoresis, nucleic acid purification (plasmid and genomic DNA, RNA), nucleic acid quantification, DNA restriction digestion and analysis, polymerase chain reaction (PCR), and basics computer-based DNA sequence analysis and data acquisition over the internet. In addition, you will learn about the nature and selection of DNA cloning vectors, restriction enzymes and other reagents used in molecular biology techniques.

Course syllabus:

Week	Dates	Lab Topics
1	Oct. 18 – 21 / 2020	Lab1: Introduction: lab rules and biosafety & DNA Isolation I
2	Oct. 24 – 28 / 2020	Lab 2: DNA Isolation II
3	Nov. 1 – 4 / 2020	Lab 3: RNA Isolation
4	Nov. 8 – 11 / 2020	Lab 4: Plasmid Isolation
5	Nov. 15– 18/ 2020	Lab5: Gel Electrophoresis & DNA Quantitation
6	Nov. 22– 25/ 2020	Lab 6: Polymerase Chain Reaction I
7	Nov. 29– Dec.2/ 2020	Lab 7: Polymerase Chain Reaction II
	Nov. 29– Dec.2/ 2020	Mid Term Exam
8	Dec. 6 – 9 / 2020	Lab8: Restriction Endonucleases
9	Dec. 13 – 16 / 2020	Lab9: Primer Design + SNP Genotyping By RFLP
10	Dec. 20 – 23 / 2020	Lab10: DNA Sequencing I
11	Dec. 27 – 30/ 2020	Lab11: DNA Sequencing II
12	Jan. 3 – 6/ 2021	Lab12: DNA Sequencing III & Transformations + Cloning
	Jan. 10 – 14/ 2021	Final Exam

Micropipette

Very important tool in almost genetics lab, for transferring or measuring minute amounts of fluid, microorganisms, etc.

There are many different volumes :

❖ **In a 200-1000 μ L micropipette: (Blue)**

The first digit is thousands.
The middle digit is hundreds.
The third is tens.

Tips colors	Volumes (μ L)
Blue	200 - 1000
yellow	2 - 200
red	<10

❖ **In a 20-200 μ L micropipette: (Yellow)**

The first digit is hundreds.
The second is tens.
The third is units.

YouTube video for more explanation:

https://www.youtube.com/watch?v=uEy_NGDfo_8

❖ **In a 0.5-10 μ L micropipette: (Red)**

The first digit is tens
The second is units
The third digit is tenths. "اعشار"

Range: **0.5-10 μ L**



Example volumes:



1.3 μ L **8.5 μ L**

Range: **2-20 μ L**



Example volumes:



3.0 μ L **17.4 μ L**

Range: **20-200 μ L**



Example volumes:



25.0 μ L **138.0 μ L**

Range: **200-1000 μ L**



Example volumes:



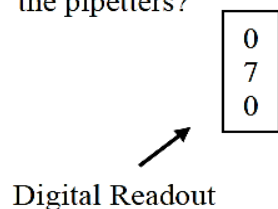
545 μ L **1000 μ L**

The red number indicates decimal on P10 and P20. The P1000 should show a red **1** ONLY when the other numbers are 0 (in other words, set at 1000 μ L).
Always hold the micropipet in a **vertical position** when there is fluid in the tip.

Practices:

1. Answer the following:

In the boxes below, write how many μL the following digital readout correspond to in each of the pipettors?



P10

P200

P1000

Write volume in μL

Which micropipettor would be appropriate to measure $250\mu\text{L}$? _____ Fill in the numbers that should appear in the digital display if that pipettor were to measure $250\mu\text{L}$.

2. Complete the table:

Volume to add to the tube	Which micropipette was used?	Record the setting as it appears in the window
400 μL		
150 μL		
850 μL		

Another type of pipette:

PIPETTE PUMP

- Easy one hand operation.
- Rotation of the thumbwheel draws or dispenses the liquid with precision and ease. By depressing the side lever the entire contents can be dispensed rapidly.
- Accepts glass and plastic pipettes.
- Color coded for sizes.



Solution

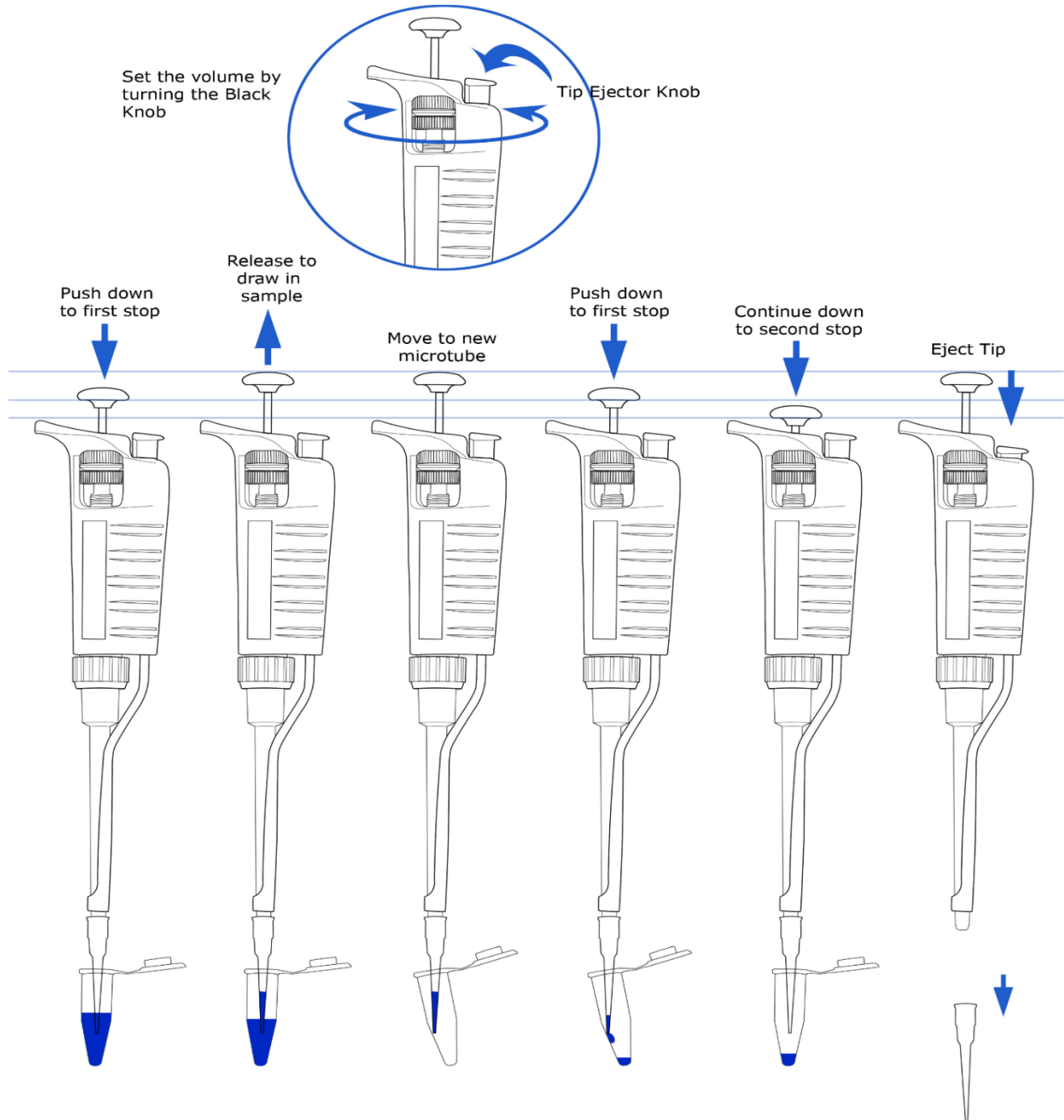
1)A) P10 = 7,0
 P200 = 70
 P1000 = 700
 B) In 200-1000 μL micropipette (Blue)
 Write as: 025

2. 400 – blue – 040

150 – yellow – 150

850 – blue – 085

How to use it?



 المايكروباييت اهم اداة في مختبرات البيولوجيا الجزيئية لازم نتقن ضبطه اذا ما تعلمتوه في هاد اللاب رح تعاونوا في باقي اللابات وحتى في الشغل.
 وييجي مرات اسئلة بالكويزات عليه.

Lab #1: Nucleic Acid Isolation

Why we isolate nucleic acid DNA or RNA?

To study the genetic causes of disease and for the development of diagnostics and drugs. It is also essential for carrying out forensic science, sequencing genomes, detecting bacteria and viruses in the environment and for determining paternity.

Briefly: “genotyping SNPs and mutation”

Remember:
SNPs more than 1%
Mutation less than 1%

Action plan in molecular lab:

1) Nucleic acid isolation:

DNA: Bacteria (*E.coli*, gram negative) , Human blood cell .

RNA: Virus , Bacteria .

2) Quantification and nucleic acid analysis:

There are many technical methods:

- 1- Spectrophotometer: Protein contamination.
- 2- Gel electrophoreses: Nucleic acid contamination.
- 3- Nano drop: Calculate N.A concentration.

3) PCR :

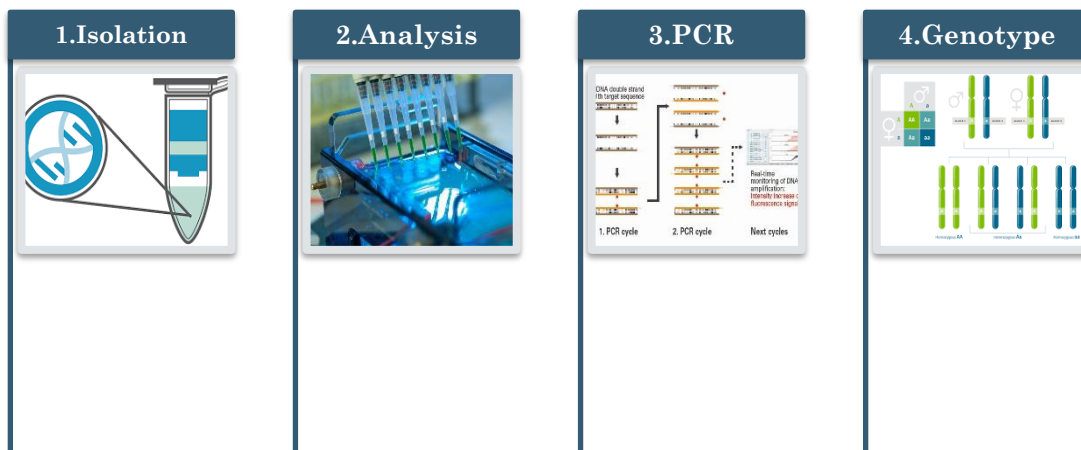
Rapidly make millions to billions of copies of a specific DNA sample

1. Long PCR
2. RT-PCR
3. Nested PCR

4) Genotype:

Method: 1) Allele specific PCR

- 2) RFLP.
- 3) Cloning.
- 4) Sequencing.
- 5) Data analysis.



Nucleic Acid Isolation (DNA)

Isolation separate to: Extraction and purification

- **Extraction:** Frees DNA from the cell and then separates it from cellular fluid and proteins so you are left with DNA molecule.
- **Purification:** Remove any remaining unwanted material and cellular debris.

- استخراج المادة الوراثية من الخلية ومن المحيط الخلوي المتواجدة فيه .
- التأكد من عدم وجود حطام خلوي يسبب تلوث للعينة .

خطأ علمي اذا قمنا بتسمية العملية باكستراكشن لوحدها لان العملية تفصل بغياب البيوريفيكيشن .

3 Major steps in DNA isolation:

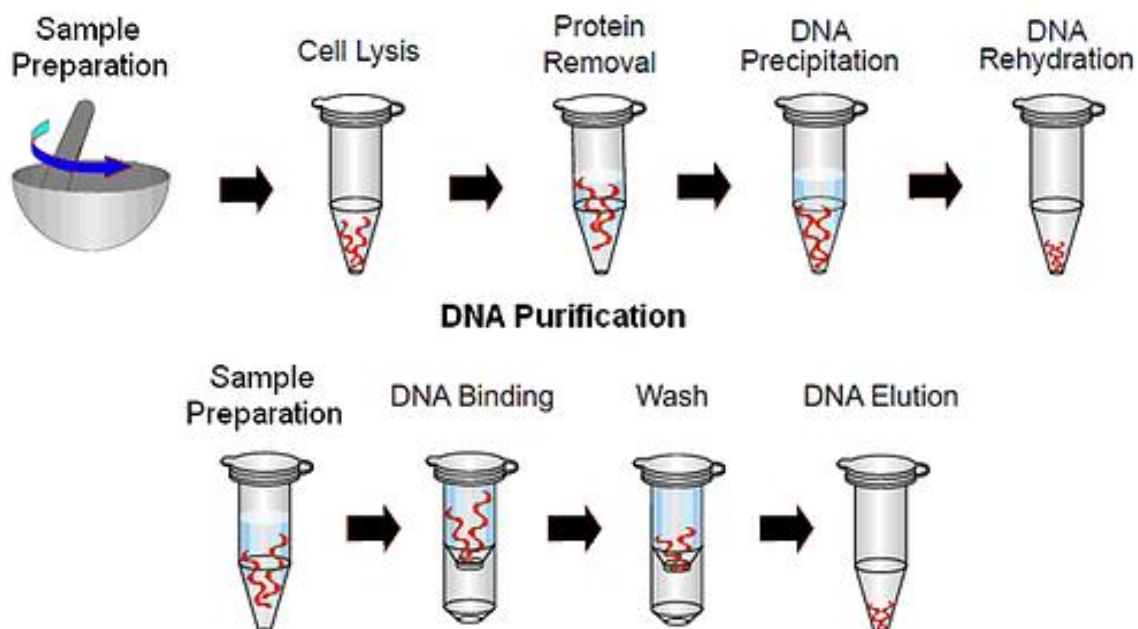
1. Cell lysis.

- ✓ Remove:
- ✓ Cell membrane and Cell wall when found.
- ✓ Cytoplasm.
- ✓ Nuclear membrane.
- ✓ Proteins, polysaccharides & lipids.

2. DNA precipitation.

When you complete the lysis step, the DNA is freed from the nucleus, but it is mixed with mashed up cell parts.

3. DNA purification



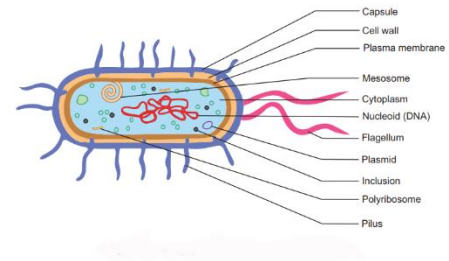
EXERCISE 1.1: DNA isolation from *Escherichia coli*

By Differential solubility:

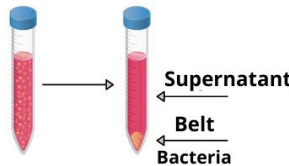
✓ Organic solvents

Salting-out

Procedure:



1. Grow *E. coli* culture overnight in broth.
2. Transfer 1.0 ml to a 1.5-ml micro centrifuge tube and spin 2 min.
(cell harvesting)



3. Decant the supernatant.
4. Resuspend the pellet in 460µl **TE buffer** by repeated pipetting.

TE buffer: Tris – EDTA

✓ Tris maintain PH = 8 | * optimum PH for DNA.

✓ EDTA (Ethylene diamine tetra acetic acid):

Inhibit DNAase, by being a chelating agent chelate divalent ions
Mg+2 (co-factor, for all nucleases) to protect DNA from degraded.

حاليا عنا الجزيء بدون انزيمات الخلية التي تعمل على تكسيره

5. Add 30 µl of 10% SDS and 10 µl of 20 mg/ml **proteinase K**, mix, and incubate 1 hr at 37 ° C.
- ✓ **NaCl**: Stabilised DNA and remains as a double helix.
 - ✓ **SDS**: which disrupts the lipid layers, helps to dissolve membranes & binds positive charges of chromosomal proteins (*histones*) to release the DNA into the solution.
 - ✓ **Proteinase K**: to digest the proteins, deproteinization of DNA and inactivation of DNase
- * **elevated temperature** (Also 56°C to inhibit degradation by DNases).

EDTA
يمنع تكسر الحمض
النووي عن طرق عزل
ايونات المغنيسيوم التي
تعتبر العامل المساعد
لهذه الانزيمات .

✓ ايون الصوديوم بحفظ جزيء ال DNA على شكل حلزوني بدون ما ينفك تهيأ لفك بروتين الهستون عنه لانه اذا صار Single strand رح يتكسر وما استفيد.
ال SDS بحطم طبقات الليبد لما يرتبط مع الهستون بسبب اختلاف الشحنة يعني بعمل "unfolding" حتى يتفكك ويطلع ال DNA بالمحلول
✓ Proteinase K: يشتغل على موضوع الروابط الببتيدية وكملّي تكسير البروتين وبتأكدلي من موضوع ال DNases.

6. In the fume hood, add an equal volume of **Phenol/chloroform/isoamyl - alcohol** and mix well but very gently to avoid shearing the DNA by inverting the tube until the phases are completely mixed.

Caution: phenol causes severe burns.

This Step is purification by “Phenol, chloroform, isoamyl alcohol”

Phenol: degrade protein 25 : 24 : 1

Chloroform: remove lipids and carbs + dissolve “phenol and Iso “

Isoamyl alcohol: anti-foaming

Name of this method is Organic solvents

هيك بنكون خلصنا من الكونتامينيشن المحيطة بالحمض النووي

7. Carefully transfer the DNA/phenol mixture and spin at 12,000 RPM for 10 min.

Aqueous phase (DNA, H₂O)

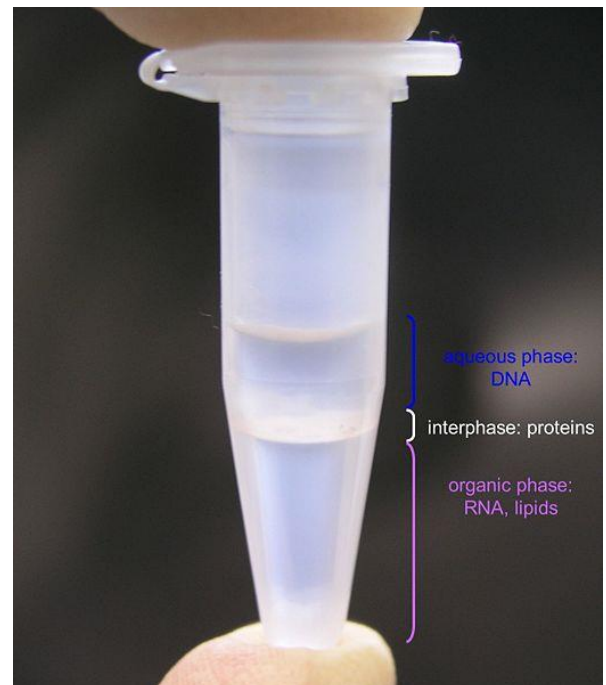
Interphase (cell debris, RNA)

Organic phase (protein, lipids)

إذا لمس التيبس بالانترفيز بدنا نرجع نعيد السيل لايسز

لانه اجي

RNA عشان هيك هاي ادق خطوة.



8. Transfer the upper aqueous phase to a new tube and add an equal volume of Phenol/chloroform/isoamyl alcohol in the fume hood.
9. Again, mix well and spin at 12,000 RPM for 10 min.

{8+9 is **Extra purification.**}

10. Transfer the upper aqueous phase to a new tube.

11. Add 1/10 volume of sodium acetate. Mix.

How can you calculate sodium acetate?

If we have 100ml aqueous you divide it by 10 → 100 / 10 = 10 ml of sodium acetate.

Function of it “stabilization of DNA”

12. Add 0.6 volumes of isopropanol and mix gently until the DNA precipitates.

Isopropanol “DNA precipitation by removing water from phosphate”

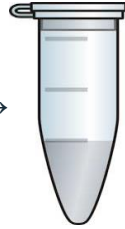
How can you calculate isopropanol?

We have ,6 volume $\rightarrow ,6 * 100 = 60 \text{ ml}$

Now we have:

H₂O-Isopropanol above $\rightarrow$

DNA-Na below $\rightarrow \rightarrow \rightarrow \rightarrow$



13. Centrifuge at 12,000 RPM for 2 min.

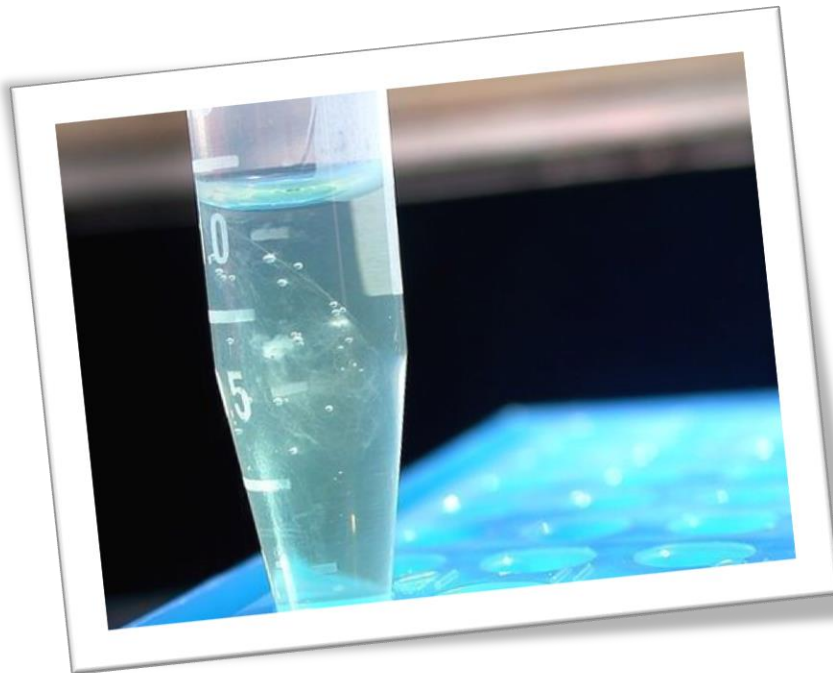
14. Decant the supernatant, add 1 ml of 70% ethanol, mix and centrifuge 12,000 RPM for 2 min.

“dissolve isopropanol + washing Na “

15. Aspirate the ethanol and air-dry the pellet (10–15 minutes)

16. Rehydrate DNA in 100 μl TE buffer. Complete resuspension may take several days. Store DNA at 4 ° C for short term, -20 or -80 ° C for long term.

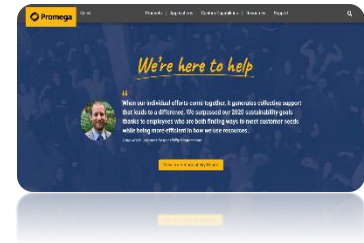
Our DNA



EXERCISE 2: DNA isolation from blood (Promega Kit)

من أهم الشركات في مجال تصنيع الكيت : *Promega Kit*

نفس المبدأ السابق لكن بوجود اختلافات صغيرة بسبب نوع العينة المراد الاستخراج منها



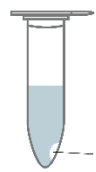
By Differential solubility:

✓ **Salting-out**

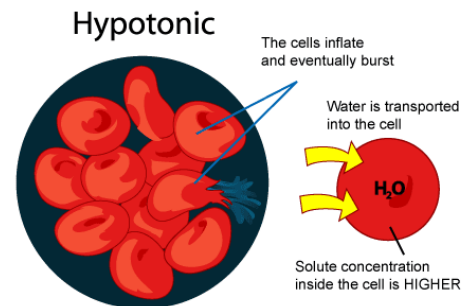
Organic solvent

Procedure:

1. Combine 900 μ l of **Cell Lysis Solution** and 300 μ l blood. Mix by inversion.
It depend on **salting out** “salt concentration” “سبب التسمية”
cell lysis is TE and Salt; salt make solution hypotonic solution for red blood cell that mean the direction of movement is from outside to inside .



يعني بالمختصر بحتاج هاد الميثود عشان يخلي الخلية تتعبي مي وتتفجر , وهيك بنكون كسرنا الغلاف الخلوي اول مرحلة لاستخراج المادة الوراثية .
طبعاً لما تتفجر بتروح ع السوبرناتنت الحطام وبترسب Nuclei ←



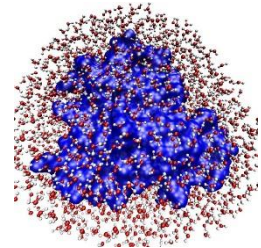
2. Incubate for 10 minutes at room temperature.
3. Centrifuge at 13,000–16,000 for 1 min.
4. Discard supernatant. Vortex pellet.
5. Add 400 μ l **Nuclei Lysis Solution** and mix by inversion.
Also make solution hypotonic but now to disrupt nuclei: so that I need salt to stabilizing DNA helix and make solution hypotonic. And disrupts the lipid layers, helps to dissolve membranes & binds positive charges of chromosomal proteins (*histones*) to release the DNA into the solution.

عشان اكسر النوية والهستون ويطرح DNA

6. Add 100 µl **Protein Precipitation Solution**; vortex for 20 seconds.

- ✓ Used to downstream products “Protein and lipid”
- ✓ Hypertonic solution
- ✓ H₂O make protein neutralization by (H⁺ and O⁻) since they associated with hydration shell and negative /positive amino acids the protein start precipitation.

هنا بالطبيعة يتكون كل مادة بمحلول محاطه بغلاف مثلا البروتين محاط بغلاف من مادة السليوشن "المذيب" بحافظ عليه بمنع تجمعه مع البروتينات الثانية (الاحماض الامينية سالبة الشحنة نظرا لارتفاع تركيز الايون بشكل غير طبيعي الهيدروجين بالاحماض الامينية سالبة الشحنة نظرا لارتفاع تركيز الايون بشكل غير طبيعي حول البروتين كونه هايبرتونك سليوشن بتكسر الغلاف ولما ترتبط الاكسجينات بالاحماض الامينية موجبة الشحنة بتكسر الغلاف ايضا , فيتجمع مع بعض البروتينات وبترسن .



7. Centrifuge at 13,000–16,000 for 3 min.

8. Transfer supernatant to a new tube containing 600 µl isopropanol

9. Centrifuge at 13,000–16,000 for 2 min.

10. Discard supernatant. Add 70% ethanol (same volume as isopropanol).

11. Centrifuge as in Step 9.

12. Aspirate the ethanol and air-dry the pellet (10–15 minutes).

13. Rehydrate the DNA in 100 µl of DNA Rehydration Solution for 1 hour at 65°C or overnight at 4°C.

 **Result**



EXERCISE 3.1: RNA isolation from Fresh blood sample

RNA is made from DNA so why we have procedure to isolation RNA without satisfied with DNA isolation?

To get some information we can't get it at DNA level like:

- ✓ **Sequence:** Estimate sequence of amino acid
- ✓ **Length of m RNA:**
 1. Number of base paring
 2. Alternative splicing
 3. More than one protein
- ✓ **Quantity of mRNA:** pattern of expression
 1. Tissue specific gene
 2. Developmental genes
- ✓ **Some viruses we can't deal with them without RNA isolation.**

Tissue specific gene: are those expressed at higher levels in a subset of tissues relative to the baseline expression all tissues.

مستوى التعبير لها اعلى في اماكن من غيره. يعني موجودة بس باختلاف نسبة الاكسبرشن.

Developmental genes: off in all cell lines except in embryonic stage

يعني بس يحتاجها لما يكون امبريو وبعدين بتضل اوف

RNA isolation also using **organic solvent** method:

General steps:

- ✓ **DEPC “Diethyl pyro carbonate”**

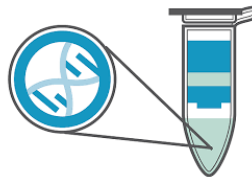
Inactivate RNase by modification histidine to mask the active site of it . Histidine is an α -amino acid that is used in the biosynthesis of proteins.
اصعب عدو للايزوليشن هو انزيم التفسير ف بخلص منه عطول عشان اضمن انه الجزيء بامان.

- ✓ **Denaturation for DNA :**

Add acidic P.C.I to make PH = 4.5

why 4.5 exact? because this is an optimum temperature for RNA.

- Supernatant: RNA
- Belt: DNA



- ✓ **Rehydration**

By RNase free water “deionized water”

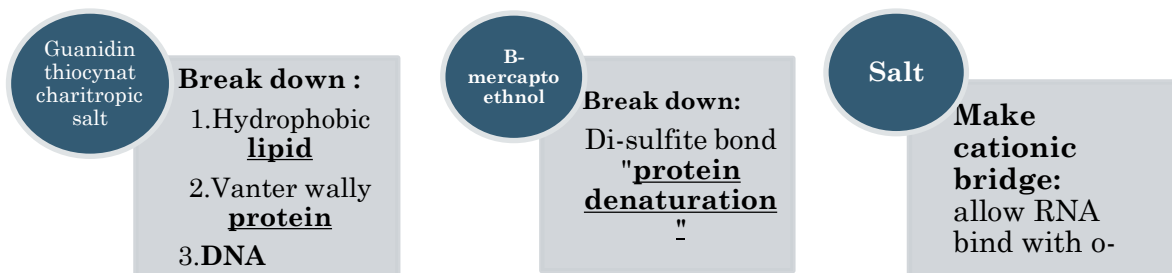
Procedure:

1. Transfer 1 ml blood to a 1.5-ml micro centrifuge tube and spin 2 min. Decant the supernatant.

Cell harvesting.

2. Suspend the pellet in 1.5 ml **EL buffer** by pipetting 4–5 times
Erythrocyte lysis buffer “RBC Lysis Buffer”
Belt: WBC precipitate.
3. Spin for 2 min. Remove 1ml of supernatant by pipetting from the top and discard. Leave the remaining supernatant and cell pellet in the tube.
4. Repeat Steps 2 and 3
5. Discard supernatant. Add 175 µl of **RNA Lysis Buffer (RLA)** buffer to pelleted leukocytes. Mix thoroughly by inversion.

RNA purification



6. Add 350µl **RNA Dilution Buffer (RDA, blue).** Mix by inverting 3–4 times
7. Place in a water bath or heating block at 70°C for 3 minutes.
8. Centrifuge for 10 minutes. Transfer the cleared lysate to a fresh tube
9. Add 200µl 95% ethanol to cleared lysate and mix well (pipet).
10. Transfer mixture to Spin Basket Assembly and centrifuge for 1 minute. Discard eluate.

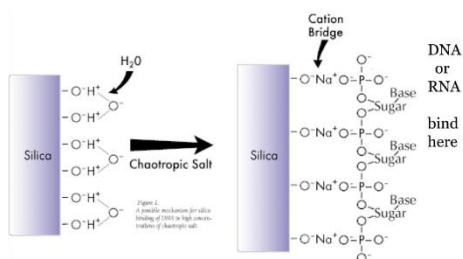
ملاحظة صغيرة:

Names of buffers and columns might differ between different kits due to different producers.

القصة كالتالي:



1. أولا بدي اعرّفكم على بطل القصة هون :
الكولوم الموضح بالشكل عبارة عن جزئين , الاصغر فيه فلتر حتى يمنع ل RNA ينزلوا ويسمح ل باقي الحطام ينزل.
الاكبر عبارة عن كولكتف بجمع الاشياء اللي ما بدي اياها عشان اكبها .



2. الانبوب الاصغر Spin column مصمم بحيث انه الجدار عبارة عن سيليكات مرتبطة مع ايون الاكسجين بشكلوا مع بعض (silanol group) , طبعا بفيديني كثير لاني بدي اخلي RNA يرتبط ويثبت واكب collective tube طبعا ما بقدر RNA يرتبط مباشره مع silanol group لا بدي وسيط , اللي هو ايون الصوديوم الموجب اللي جاي من الملح اللي انضاف مع buffer حتى يرتبط مع ايون الاكسجين السالب ويعمل cation bridge , بهاي الحالة يكون ارتبط RNA .

Silanol group = Si - O

Cation bridge = Si - O ions - Na ions

✓ Si - O ions - Na ions - RNA

3. مشكلة صغيرة !

مرات بكون ضايل في DNA موجود وبروتينات (اللي بهمني منهم ال negative) يرتبطوا معه بدل ال RNA بال cation bridge طبعا نسبتهم قليلة ومرات في H₂O , عشان احلها وتضل عينتي بس RNA نقية مافيهها contamination منهم بعمل **WASH**

11. Add 600µl of **RNA Wash Solution (RWA)**. Centrifuge for 1 minute and discard the eluate.
12. Apply 50µl of DNase mix to membrane. Incubate at RT for 15 minutes.
13. Add 200µl **DNase Stop Solution (DSA)** and centrifuge for 1 minute
14. Add 600µl **RNA Wash Solution (RWA)**; centrifuge for 1 minute. Empty
15. dd 250µl **RNA Wash Solution (RWA)**; centrifuge for 2 minutes.

Washing buffer contain (guanidine thiocyanate + isopropanol)

كل ما عملت غسل اكثر بتزيد البيوريتي اكثر بس ما بتوصل ل 100%

16. Transfer the spin column into a 1.5 ml microcentrifuge tube and add 100µl **Nuclease-Free Water** to membrane. Centrifuge for 1 minute to elute the RNA and store at -70°C.

Elute step:

Break down cation bridge by dissolve Na⁺ to elute RNA to Eppendorf tube.

If isolation of mRNA is needed:

As we know we have 3 type of RNA:

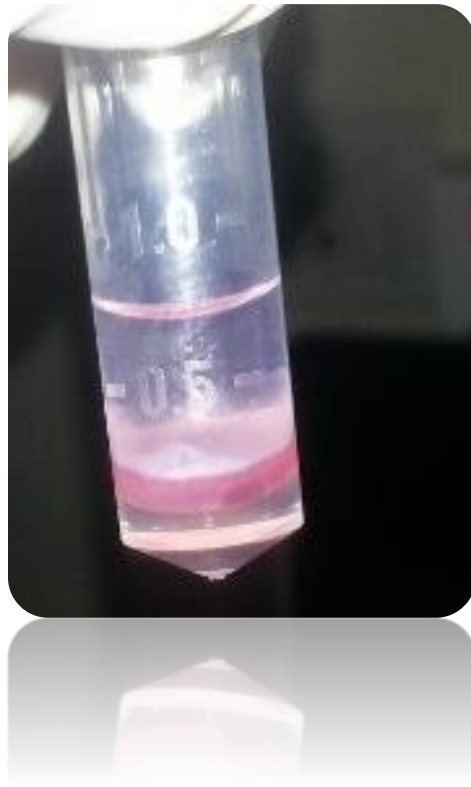
1. M-RNA
2% - 5% exist in cell (the least possible)
2. T-RNA
3. R-RNA
>90% the highest; because its associate with all small structures

We will:

Design strand of oligo dT to bind with A tale of M-RNA.

- ✓ M-RNA is bind
- ✓ T-RNA and R-RNA discard

Our RNA results:



*تجربة Plasmid DNA isolation from *E.coli* لاحقاً.

Quiz number 1

Q 1: Mention three precautions used to inactivate RNase enzyme during RNA isolation procedure:

Q 2: During RNA isolation procedure the purpose of using the following material is:

- Spin column:
- RNase- free water:

Gel Electrophoresis

Laboratory technique used to **separate** and **purify** macromolecules “Nucleic acids and Protein”.

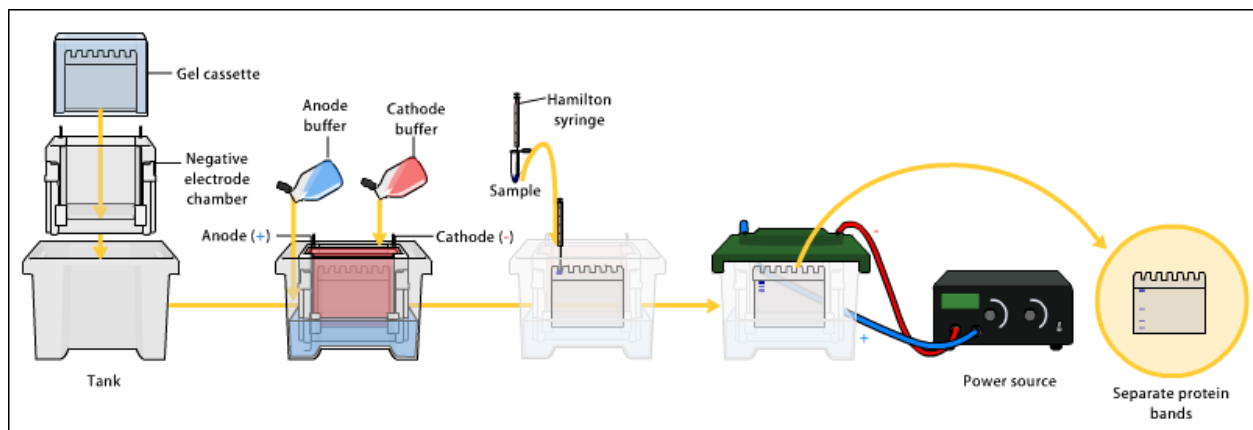
✓ Types of gel electrophoresis:

- Agarose
1D - horizontal.
- Polyacrylamide PAGE
2D - vertical

لاب المول في جامعتنا رح نحكي عن
agarose بكل مفصل , لكن انا رح احط
polyacrylamide عن
لانه مهم وبننتظرقله كثير .

- **Nucleic acid molecules** are separated by applying an **electric field** to move the negatively charged molecules through a matrix of **agarose gel**.
* briefly separate by size “faster is smaller” because all of them are negative charge.
- **Proteins** are separated by the **charge** and **size** in **polyacrylamide gel** also we can separate them in **agarose gel** by **charge** because the pores of the gel are too small to sieve proteins.
- **Most modern DNA separation methods now use agarose gels, except** for particularly **small DNA fragments use polyacrylamide gel** because it can separate DNA fragment even if differ (1 to 5) base pair unlike agarose gel that separate them from each other after 10 b.p.

SDS-PAGE-Electrophoresis:

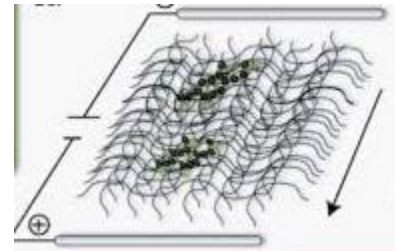


Exercise 4.1: Agarose Gel Electrophoresis

- ✓ **Agarose:** A porous material derived from red **seaweed**, is a chain of sugar molecules.
- ✓ **Agarose gel electrophoresis:** is one of several physical methods for determining the size of DNA.

- ✓ **Principle:**

DNA is forced to migrate through a highly cross-linked agarose matrix in response to an electric current. In solution, the phosphates of the DNA are negatively charged, and the molecule will therefore migrate to the positive pole.

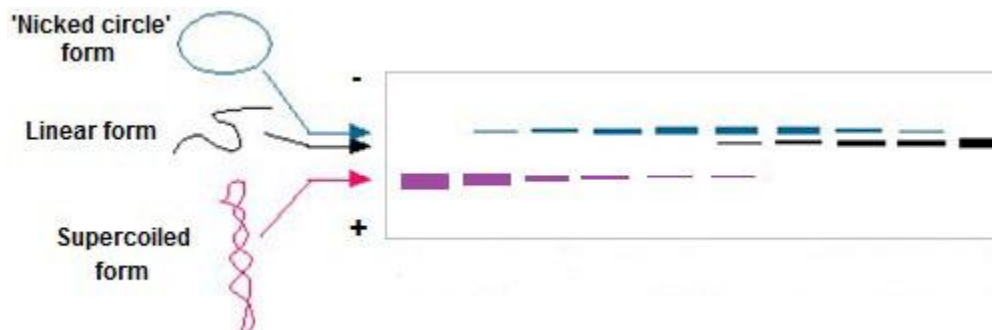


- ✓ **There are factors that affect rate and migration through a gel:**

1. **Size of the DNA:**

Smallest move fastest

2. **Shap of the DNA:**



لازم يكونوا يا كلهم نفس عدد القواعد النايتروجينية يا نفس الشكل ما بيزبط الاشيين يكونوا مختلفين بنفس الوقت

3. **The strength of the electrical field**

Increase in voltage = Increase in rate (طبعاً ضمن حدود)

4. **The state of hydration of the DNA**

which is stabilised by hydrogen interactions between base pairs
depend on conformation of DNA (A, B and Z forms)

5. **The viscosity of the solution لزوجة المحلول**

6. **The pore size of the gel.**

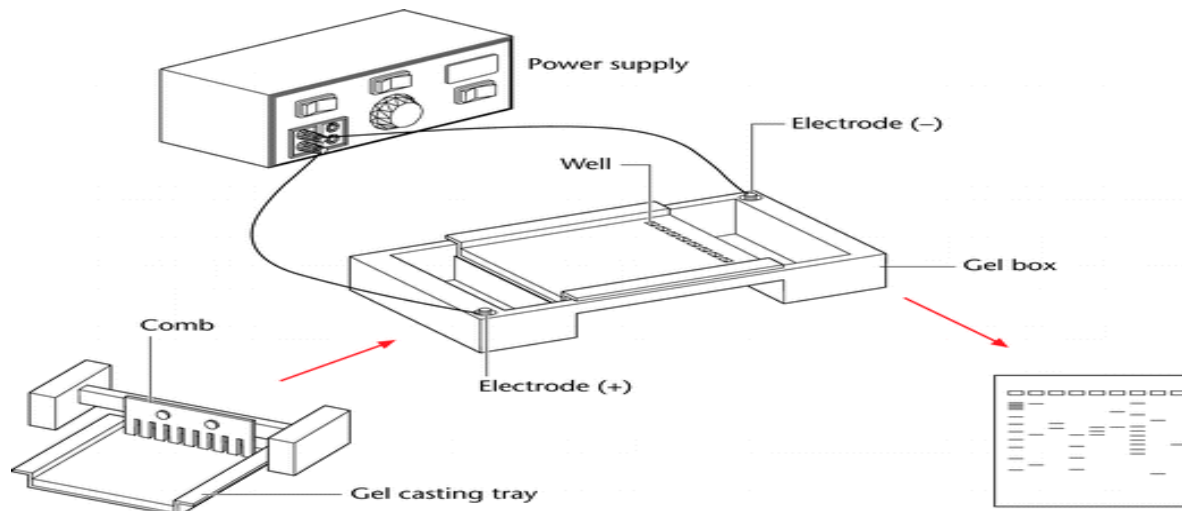
7. **DNA stains**

✓ **Application:**

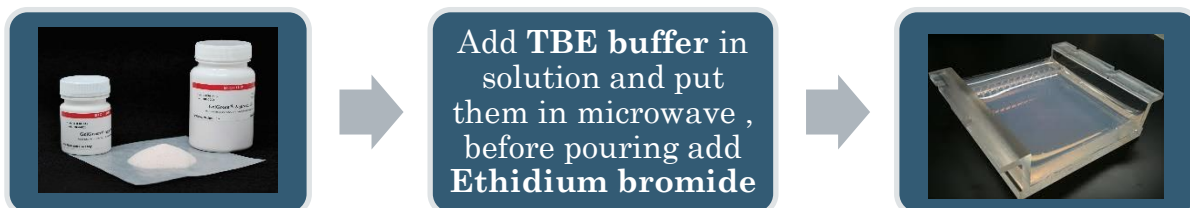
1. Estimation of the size of DNA molecules following digestion with restriction enzymes, **for example** in restriction mapping of cloned DNA.
2. Analysis of products of a polymerase chain reaction (PCR) **for example** in molecular genetic diagnosis or genetic finger printing.
3. Separation of DNA fragments **for** extraction and purification.
4. Separation of restricted genomic DNA **before** Southern transfer, or of RNA **before** Northern transfer.
5. Separation of proteins, **for example**, screening of protein abnormalities

✓ **Procedure:**

1. Prepare gel tray and place comb in gel tray.



2. Prepare a 1 % agarose solution, measure 1.0 g agarose into a flask and add 100 ml 1X **TBE buffer**. Microwave **until agarose is dissolved and solution is clear**.
3. Allow solution to cool to about 60°C **before pouring**. Add 1 µl of **Ethidium bromide**.
4. Pour 60°C gel solution into tray and allow gel to **solidify** about **30 minutes** at room temperature.



❖ **Agarose percentage:**

Low [agarose] = larger pores → better resolution of larger DNA fragments

High [agarose] = smaller pores → better resolution of smaller DNA fragments

*يعني كل ما زدنا اجاروز بنكون رصينا وصغرنا الفراغات فلما يكون عندي قطع صغيره بفيديني عشان اشوف كل قطعه على شكل band وفي جدول بالسلاید بوضح اكثر الفكرة .

❖ **TBE buffer & TAE buffer:**

As we know:

- ✓ Tris: Maintain the pH 8
- ✓ EDTA: Inactivation nucleus enzyme.

But here also we need specific buffer includes standard fit with our experiment purpose to:

- ✓ Determine the electric charge on the solute
- ✓ Carries the applied current

Tris-Borate-EDTA(TBE): Greater buffering capacity and gives better resolution with shaper bands.

Tris-Acetate-EDTA(TAE): Lower buffering capacity and mainly used when DNA are purified from the agarose gels.

الوظائف المعروفة
من التجارب السابقة
لل buffer
انه يحافظي على
الجزء من عدة
عوامل لكن هنا
بنحتاج buffer
يكون قادر يحدد
الشحنة الكهربائية
"السالبة في
تجربتنا" على
القطع.
وقادر ينقل التيار
الكهربائي حتى
يصير
ال migration

Solution:

Agarose (g)	TBE (ml)
0.5	50
1	100
1.5	150
2	200

بنستخدم قانون التخفيف حتى اعرف
الكميات تماما اللي بتعطيني حجم مناسب
لتجربتي .

$$C1*V1=C2*V2$$

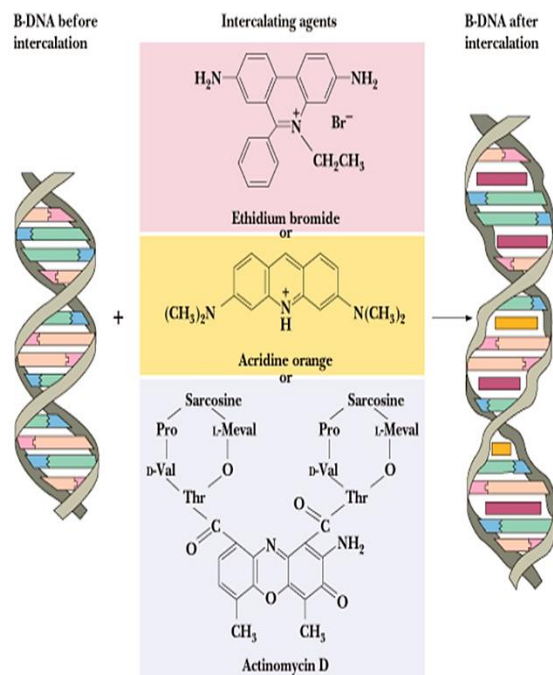
❖ Ethidium bromide (DNA stain)

* Several hydrophobic molecules containing flat aromatic and fused heterocyclic rings **can insert between the stacked base pairs of DNAs** "intercalating agents".

* لانه الصبغة الها خصائص ومكونات فتناسبت مع وظيفتها بالصبغ
عن طريق دخولها بين القواعد النايروجينية وتصبغهم عشان نشوفهم
تحت الاشعة فوق بنفسجية.

* Act as a mutagen because it intercalates double-stranded.

* بهالحالة القطعة بتخرب خلص يعني لدرجة اعمل عليها كمان تجربه
بزيطش , تخيل لو انه في جسم الانسان وتعرض للصبغة رح يصير في
سرطانات وامراض.
*الصبغة لونها احمر.



- To run, gently remove the comb, place tray in electrophoresis chamber, and cover (just until wells are submerged) with electrophoresis buffer (the same buffer used to prepare the agarose).
- To prepare samples for electrophoresis, add 2 µl of 6x gel **loading dye** for every 5 µl of DNA solution. Mix well & load. Don't forget to load the **DNA size marker**.

Loading Dye

1. Dense material:

Allow the sample to "fall" into the sample wells. تكون ثابتة ما تطلع برى النفق المخصص (e.g. glycerol or sucrose)

2. Tracking dyes:

Migrate in the gel and allow visual monitoring or how far the electrophoresis has proceeded. (e.g. xylene cyanol / bromophenol blue)

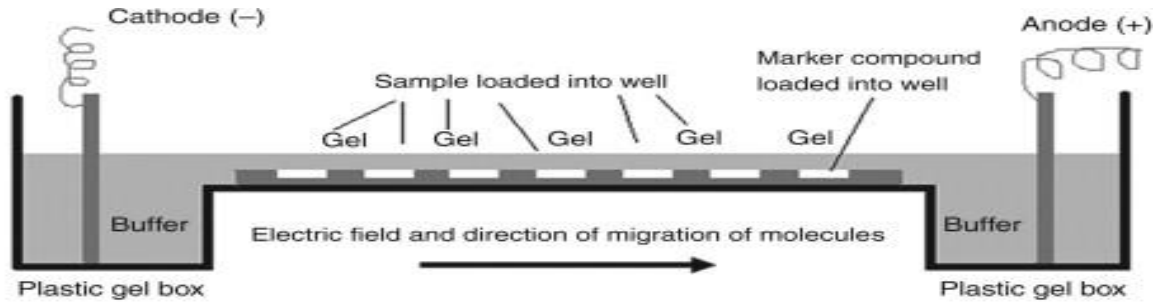
*بتسمحلها تمر مع التيار الكهربائي واحنا شوف مرورها تحت الاشعة.

*بحط بتجربتي DNA + loading dye

فبتكون كتلتها قليلة وشحنها سالبه واسرع من العينات اللي عندي عشان لما توصل لآخر القطب الموجب انتبه لعينتي انها ما تطلع برا الجل ولما اشوف تحت الاشعة القى نتائج فهي كانت عينة حقيقية لكن بتكون مؤشر .

*اذا ما لقينا نتائج:

يا نسينا نصبغ يا خلىنا التيار الكهربائي وقت اطول يا غيرنا وصل الاقطاب "الخريطة بالاسلاك".



a. Side view of gel electrophoresis

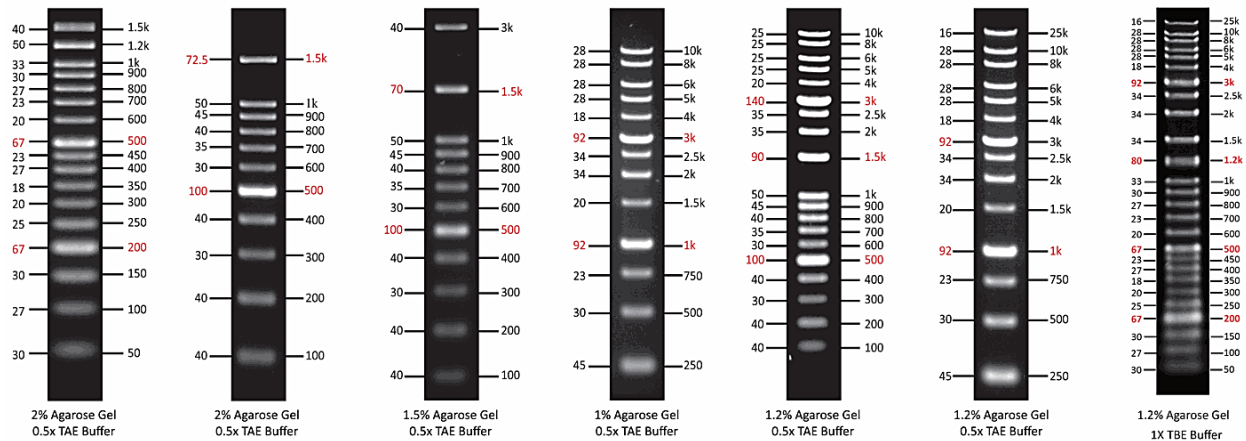


b. Top view of gel electrophoresis



2. DNA markers and ladder:

Mixture of DNA fragments of known lengths making them suitable for estimating the fragment length of concurrently run samples.



باللابات بنشترية بهاد الشكل باختلاف الشركة طبعا:

<https://www.civicbio.com/selection-tools/dna-ladder-selection-guide/>

1-877-661-4711 info@civicbio.com PCR Success Selection & Comparisons Protocols Latest News My Account 0 Items

Choose the most suitable DNA ladder for agarose gel electrophoresis.

Trans2K® DNA Marker (bp): 1,000, 750, 500, 250, 100. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

Trans2K® Plus DNA Marker (bp): 1,000, 750, 500, 250, 100. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

Trans2K® Plus II DNA Marker (bp): 1,000, 750, 500, 250, 100. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

Trans5K DNA Marker (bp): 15,000, 10,000, 7,500, 5,000, 3,000, 1,500, 1,000, 500, 250, 100. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

Trans15K DNA Marker (bp): 15,000, 10,000, 7,500, 5,000, 3,000, 1,500, 1,000, 500, 250, 100. 0.8% TAE agarose gel stained with GelStar (loading volume: 6 µl).

1Kb DNA Ladder (bp): 10,000, 8,000, 6,000, 5,000, 4,000, 3,000, 2,000. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

1Kb Plus DNA Ladder (bp): 10,000, 8,000, 6,000, 5,000, 4,000, 3,000, 2,000, 1,500, 1,000, 800. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

100bp DNA Ladder (bp): 1,000, 900, 800, 700, 600, 500. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

100bp Plus DNA Ladder (bp): 1,000, 900, 800, 700, 600, 500. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

100bp Plus II DNA Ladder (bp): 1,000, 900, 800, 700, 600, 500. 1.6% TAE agarose gel stained with GelStar (loading volume: 6 µl).

BEST SELLERS
PRODUCTS
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PROMOTIONS

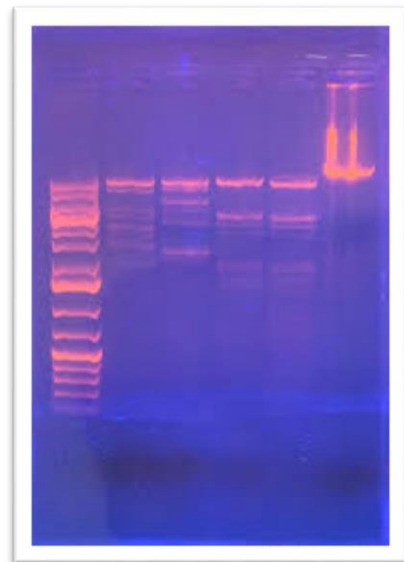
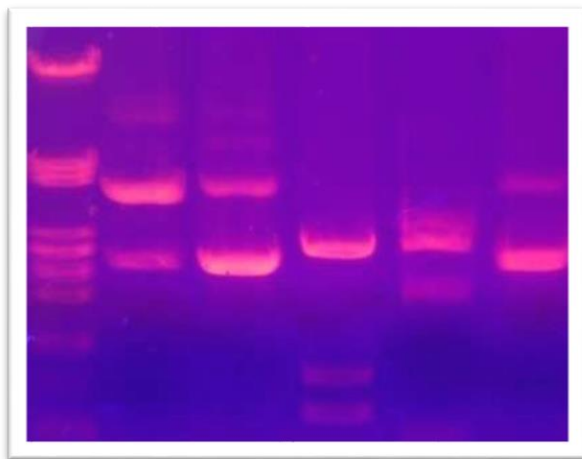
7. Electrophoresis at 50-150 volts **until dye markers have migrated** an appropriate distance, **depending on the size of DNA to be visualized**.

Power supply:

Voltage	Time
150 V	30 mins
120 V	1-1.5 hours
100 V	2-2.5 hours
90 V	3-4 hours

8. Visualize DNA under the **UV transilluminator**.

 Our result:



Quiz number 2

1. How does the size of the DNA fragment affects its movement or migration through the agarose gel during electrophoresis?
2. Predict what would happen if you forgot to add the sample loading.